Exp 10

Write the python program for Map Coloring to implement CSP.

Aim:

To write a python program for map coloring to implement CSP.

Algorithm:

Initialize an empty assignment for the map coloring problem.

For each country in the map, add a variable X_i to the assignment.

Define the constraints as pairs of adjacent countries, and add a constraint (X_i, X_j) to the CSP such that X_i != X_j.

Define the domain of each variable to be the set of available colors.

Call the Backtrack function to find a consistent assignment for the variables.

In the Backtrack function, check if the assignment is complete. If it is, return the assignment.

Select an unassigned variable X_i from the assignment.

For each value v in the domain of X_i, check if it is consistent with the constraints and the assignment so far.

If it is, assign X_i = v to the assignment and recursively call Backtrack with the new assignment.

If the result of Backtrack is not a failure, return the result.

If the result is a failure, remove the assignment X_i = v and try the next value in the domain.

If all values in the domain have been tried and failed, return failure.

Program:

```
# List of colors

colors = ["Red", "Green", "Blue"]


# Map and its neighboring regions

graph = {

    "A": ["B", "C"],
```

```
    "B": ["A", "C", "D"],
    "C": ["A", "B", "D"],
    "D": ["B", "C"]
}

# Store assigned colors
result = {}

# Function to check if a color can be assigned
def is_safe(region, color):
    for neighbor in graph[region]:
        if neighbor in result and result[neighbor] == color:
            return False
    return True

# Function to color the map
def map_coloring(region_list, index):
    if index == len(region_list):
        return True

    region = region_list[index]

    for color in colors:
        if is_safe(region, color):
            result[region] = color

            if map_coloring(region_list, index + 1):
                return True

            del result[region]
```

```
    return False


# Driver code
regions = list(graph.keys())


if map_coloring(regions, 0):
    print("Map Coloring Solution:")
    for region in regions:
        print(region, "->", result[region])
else:
    print("No solution found")
```

Output:

```
>>>
Map Coloring Solution:
B -> Red
D -> Green
A -> Green
C -> Blue
>>>
```

Result:

Hence, the simple python program for Map Coloring to implement CSP is done